Data Structures: Binary Trees and Graph Algorithms

Complete Unit 3 Lecture Notes & Practice Problems

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1. Introduction to Non-Linear Data Structures: Trees and Binary Trees.
2. Tree Traversals: Pre-order, In-order, Post-order and Level-order.
3. Binary Search Trees BST: Insertion, Deletion, and Searching complexity Oh.
4. Balanced Search Trees: AVL Trees and Rotations LL, RR, LR, RL.
5. Graph Representations: Adjacency Matrix vs Adjacency List.
6. Graph Traversals: Depth First Search DFS using Stack and Breadth First Search BFS using Queue.
7. Shortest Path Algorithms: Dijkstras Algorithm and Bellman-Ford.
8. Minimum Spanning Trees: Kruskals and Prims greedy strategies.

Practice Exercise: Implement AVL rotation check in C++ / Java.